

Initial Post

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Initial Post

by [Ali Alhammadi](#) - Wednesday, 6 August 2025, 6:44 PM

Agent-based systems (ABS) have become more popular in organizing intelligent, decentralized systems that coordinate complex and dynamic environments. Conventional systems are based on central control and fixed rules and usually do not work well in speed or scale (Axtell & Farmer, 2025). Conversely, ABS are independent agents who can perceive their surroundings, decide, and communicate with other agents to reach self- or group goals (Tour et al., 2021). The increase in ABS is deeply related to the progress in artificial intelligence. Such systems use methods of AI like machine learning, logic reasoning, and modeling behaviors. This merging enables the agents to behave independently, experience, and adapt to developments with no explicit human direction (Zolfagharipoor & Ahmadi, 2021). With the increasing size of technology and data, there has also been an increasing need to find systems that will deal with distributed processes, which makes ABS logical.

Agent-based systems bring several advantages to organizations. To begin with, they increase flexibility and scalability. Adding new agents can be done without reconfiguring the whole system, making expanding or changing operations easier. Second, they provide fault tolerance and strength, which is to say that should one of the agents malfunction, the rest can still operate. Third, AB facilitates decentralized decision-making, which minimizes bottlenecks and quick responses (Cincotti et al., 2022). Such systems can benefit supply chain management, robotics, financial markets, and healthcare, where autonomous elements must act in real time. Simulations can be performed using ABS to describe social, economic, or environmental systems, in order to understand emerging behavior and system-level outcomes. Examples of agent models include reactive agent models (responsive to instant inputs), deliberative agent models (plans using internally based goals), and hybrid models. Multi-agent systems take this further by allowing cooperation or competition among agents and resemble how human beings are organized (Rand & Stummer, 2021). In summary, ABS offers a dynamic, innovative business structure in a dynamic environment.

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Peer Response

by [Abdulla Husain Salem Hadna Almessabi](#) - Thursday, 7 August 2025, 1:10 PM

Hi Ali,

The post is a perfect introduction to ABS and its importance in a decentralised dynamic environment. I specifically like how you point out how ABS, rather than the classic systems, utilise the ability of independent agents who can not only perceive, make decisions, and interact to pursue self- or group objectives. Indeed, such flexibility is one of the fundamental strengths of the visual tool because individual organisations can work with a lot more elasticity and capacity (dos Santos & Saraiva, 2021).

Improved fault tolerance and decentralised decision-making are essential to industries that have to respond to the changes in real-time and be resilient, such as supply chain management, robotics, and healthcare (Fischbach et al., 2021). I believe that what you said about ABS supporting fault tolerance is particularly important in the present systems, where it is paramount to business continuity to ensure the system goes on regardless of the malfunctions that hit its agents. Further, its adaptability and learning capacities over time via machine learning, logical rules, and behavioural models are additional elements of sophistication that are traditionally missing in other systems (Tour et al., 2021).

You mentioned various models of agents, including reactive, deliberative, and hybrid models. The realisation that ABS can be used to facilitate cooperation among the agents depicts real-world interactions and thus the models are relevant in modelling activities of complex socio-economic systems (Turgut & Bozdog, 2023). They are used in financial markets, and your mention of them gives us useful information about market behaviours and patterns.

ABS can provide a useful framework under which the organisation would be able to make operations more efficient and achieve a strategic advantage due to decentralised decisions and versatility (Værbak et al., 2021).

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Peer Response

by [Ali Alzahmi](#) - Sunday, 10 August 2025, 12:37 PM

In your post, you present a very balanced and comprehensive introduction to agent-based systems (ABS), with its obvious beginning presented in contrast to the more commonplace standard and rigid understanding of centralised modes of systematisation. I also liked how you showed the importance of artificial intelligence methods to empower the agents to act autonomously and adjust to their world: machine learning, logical reasoning, and behavior modelling (Axtell & Farmer, 2025). Such interrelation of ABS and AI is paramount, since it reveals that even though their emergent usage is seen as a form of decentralisation, they are also about greater intelligence and adaptability. The way you addressed every single one of the

described advantages (flexibility, scalability, fault tolerance, decentralised decision-making) was quite solid. You correlated these improvements to real-life practices related to supply chains, robotics, financial markets, and healthcare, making the argument more practical (Ghaffarzadegan et al., 2024).

Another strength was your explanation of the various models of agents, reactive, deliberative, and the hybrid, and how they are used in simulations, which depicts the flexibility of ABS in the representation of complex systems. To increase the strength of your analysis, you might identify any difficulty regarding the implementation, like the impossibility of coordinating with many agents working independently or the aspects of governance relative to transparency and accountability in the decision process (Sakurada et al., 2022). Also, it can be beneficial to elaborate more on how multi-agent systems with other emerging technologies, such as IoT, cloud computing, or blockchain, might make them more efficient and trustworthy (Ghaffarzadegan et al., 2024). Generally, your answer aptly expresses the idea behind ABS's conceptual framework and practical consequences. It offers a flexible, forward-looking solution to cope with complexity in the contemporary organisational world.

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Peer Response

by [Mansour Saeed Mubarak Hamdan Al Hamdani](#) - Monday, 11 August 2025, 8:23 AM

The merit of operator-based agent systems (ABS) was analysed with a wise contrast to centralized systems. The positive advantages include flexibility, fault tolerance, and decentralized decision-making, supported extensively by Axtell and Farmer (2025) and Cincotti et al. (2022). The observation relating to AI upheaval and independent agent adaptation through deep learning fits well, given the fast-moving nature of fields like supply chain and healthcare, where environmental variables transform quickly. The reference to reactive vs. deliberative or hybrid models provides useful context for integration avenues, although the question of multi-agent cooperation remains significant (Cincotti et al, 2022). The thought that simulations of organizational behavior are important is accepted, but as you fairly observe, prioritizing the good of the whole can be tricky; especially under competition. In real-time applications, conflict resolution methods may have to integrate adaptive algorithms with predefined coordination mechanisms to avoid deadlock or counterproductive rivalry. On the other hand, the hybrid approach could benefit industries such as logistics, energy distribution, smart manufacturing, and disaster response by merging decentralized adaptability with centralized supervision stability. Overall, your analysis addresses both the potentials and unresolved operational challenges concerning ABS.

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Re: Peer Response

by [Rayyan Mohamed Abdalla Alshambeeli Alnaqbi](#) - Sunday, 17 August 2025, 7:25 PM

Ali, Your summary of agent-based systems (ABS) does a great job of showing how deep they are technically and how useful they are in the real world. I really liked how you broke down the different types of agents—reactive, deliberative, and hybrid—because it shows how smart and independent these systems can be.

To add to what you've said, one of ABS's main strengths is that they can help people make decisions in real time in very unpredictable situations. This is especially important in areas like self-driving cars and disaster response, where things change quickly and decisions need to be made right away and in person (Stone and Veloso, 2000). Here, ABS are better than older systems because they let agents organize themselves without needing central coordination.

Also, your mention of multi-agent systems (MAS) is right on the money. MAS do more than just give people freedom; they also model social behavior and negotiation, which is important in areas like smart energy grids and collaborative robotics. These agents can not only work together, but they can also model competitive settings that are similar to how organizations work in the real world (Sycara, 1998).

I would also say that ABS are useful for making long-term predictions. For instance, businesses use ABS-based simulations to see how policies will work, how the market will change, or even how false information will spread on social media (Macal and North, 2016). Because they can predict the future, ABS are not just reactive tools; they also help businesses be flexible and adaptable.

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Re: Peer Response

by [Mariam Ibrahim Ismail Hasan Almarzooqi](#) - Friday, 17 October 2025, 3:56 PM

Hi,

Thank you for your detailed and insightful post. You've clearly highlighted the growing importance of agent-based systems (ABS) in managing complex and decentralized environments. I agree with your point that traditional centralized systems often lack the flexibility and scalability required in today's fast-paced, data-driven world, and ABS presents a compelling alternative.

Your explanation of how agents operate independently, perceiving, reasoning, and collaborating, is particularly relevant to real-time applications such as supply chains and healthcare. I also appreciate your mention of different agent models (reactive, deliberative, and hybrid) and how multi-agent systems mirror human organizational behavior. This comparison really emphasizes the value of ABS in designing adaptive and resilient systems.

It's exciting to see how developments in AI, particularly machine learning and behavioral modeling, continue to enhance agent capabilities. As these systems become more advanced, I believe the challenge will lie not only in implementation but also in ensuring ethical and transparent decision-making, especially in high-stakes environments.

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